

2. ■■■■■■ — ■■■■ ■■■■

[illegible][illegible]



1. 1940-1941 2. 1942-1943 3. 1944-1945 4. 1946-1947 5. 1948-1949 6. 1950-1951 7. 1952-1953 8. 1954-1955 9. 1956-1957 10. 1958-1959 11. 1960-1961 12. 1962-1963 13. 1964-1965 14. 1966-1967 15. 1968-1969 16. 1970-1971 17. 1972-1973 18. 1974-1975 19. 1976-1977 20. 1978-1979 21. 1980-1981 22. 1982-1983 23. 1984-1985 24. 1986-1987 25. 1988-1989 26. 1990-1991 27. 1992-1993 28. 1994-1995 29. 1996-1997 30. 1998-1999 31. 2000-2001 32. 2002-2003 33. 2004-2005 34. 2006-2007 35. 2008-2009 36. 2010-2011 37. 2012-2013 38. 2014-2015 39. 2016-2017 40. 2018-2019 41. 2020-2021 42. 2022-2023 43. 2024-2025 44. 2026-2027 45. 2028-2029 46. 2030-2031 47. 2032-2033 48. 2034-2035 49. 2036-2037 50. 2038-2039 51. 2040-2041 52. 2042-2043 53. 2044-2045 54. 2046-2047 55. 2048-2049 56. 2050-2051 57. 2052-2053 58. 2054-2055 59. 2056-2057 60. 2058-2059 61. 2060-2061 62. 2062-2063 63. 2064-2065 64. 2066-2067 65. 2068-2069 66. 2070-2071 67. 2072-2073 68. 2074-2075 69. 2076-2077 70. 2078-2079 71. 2080-2081 72. 2082-2083 73. 2084-2085 74. 2086-2087 75. 2088-2089 76. 2090-2091 77. 2092-2093 78. 2094-2095 79. 2096-2097 80. 2098-2099 81. 2100-2101 82. 2102-2103 83. 2104-2105 84. 2106-2107 85. 2108-2109 86. 2110-2111 87. 2112-2113 88. 2114-2115 89. 2116-2117 90. 2118-2119 91. 2120-2121 92. 2122-2123 93. 2124-2125 94. 2126-2127 95. 2128-2129 96. 2130-2131 97. 2132-2133 98. 2134-2135 99. 2136-2137 100. 2138-2139 101. 2140-2141 102. 2142-2143 103. 2144-2145 104. 2146-2147 105. 2148-2149 106. 2150-2151 107. 2152-2153 108. 2154-2155 109. 2156-2157 110. 2158-2159 111. 2160-2161 112. 2162-2163 113. 2164-2165 114. 2166-2167 115. 2168-2169 116. 2170-2171 117. 2172-2173 118. 2174-2175 119. 2176-2177 120. 2178-2179 121. 2180-2181 122. 2182-2183 123. 2184-2185 124. 2186-2187 125. 2188-2189 126. 2190-2191 127. 2192-2193 128. 2194-2195 129. 2196-2197 130. 2198-2199 131. 2200-2201 132. 2202-2203 133. 2204-2205 134. 2206-2207 135. 2208-2209 136. 2210-2211 137. 2212-2213 138. 2214-2215 139. 2216-2217 140. 2218-2219 141. 2220-2221 142. 2222-2223 143. 2224-2225 144. 2226-2227 145. 2228-2229 146. 2230-2231 147. 2232-2233 148. 2234-2235 149. 2236-2237 150. 2238-2239 151. 2240-2241 152. 2242-2243 153. 2244-2245 154. 2246-2247 155. 2248-2249 156. 2250-2251 157. 2252-2253 158. 2254-2255 159. 2256-2257 160. 2258-2259 161. 2260-2261 162. 2262-2263 163. 2264-2265 164. 2266-2267 165. 2268-2269 166. 2270-2271 167. 2272-2273 168. 2274-2275 169. 2276-2277 170. 2278-2279 171. 2280-2281 172. 2282-2283 173. 2284-2285 174. 2286-2287 175. 2288-2289 176. 2290-2291 177. 2292-2293 178. 2294-2295 179. 2296-2297 180. 2298-2299 181. 2300-2301 182. 2302-2303 183. 2304-2305 184. 2306-2307 185. 2308-2309 186. 2310-2311 187. 2312-2313 188. 2314-2315 189. 2316-2317 190. 2318-2319 191. 2320-2321 192. 2322-2323 193. 2324-2325 194. 2326-2327 195. 2328-2329 196. 2330-2331 197. 2332-2333 198. 2334-2335 199. 2336-2337 200. 2338-2339 201. 2340-2341 202. 2342-2343 203. 2344-2345 204. 2346-2347 205. 2348-2349 206. 2350-2351 207. 2352-2353 208. 2354-2355 209. 2356-2357 210. 2358-2359 211. 2360-2361 212. 2362-2363 213. 2364-2365 214. 2366-2367 215. 2368-2369 216. 2370-2371 217. 2372-2373 218. 2374-2375 219. 2376-2377 220. 2378-2379 221. 23

■■■■■■■■■: 69 ■■■■ ■■■■■■■■ ■■ 86. ■■■■■■ ■■■ ■■■■ ■■■■■■■■ ■■■■. 10% ■■■■■■.

4. —

	0.1–2		
	15–30	7±2	
		4–7	+
	–		→

7±2

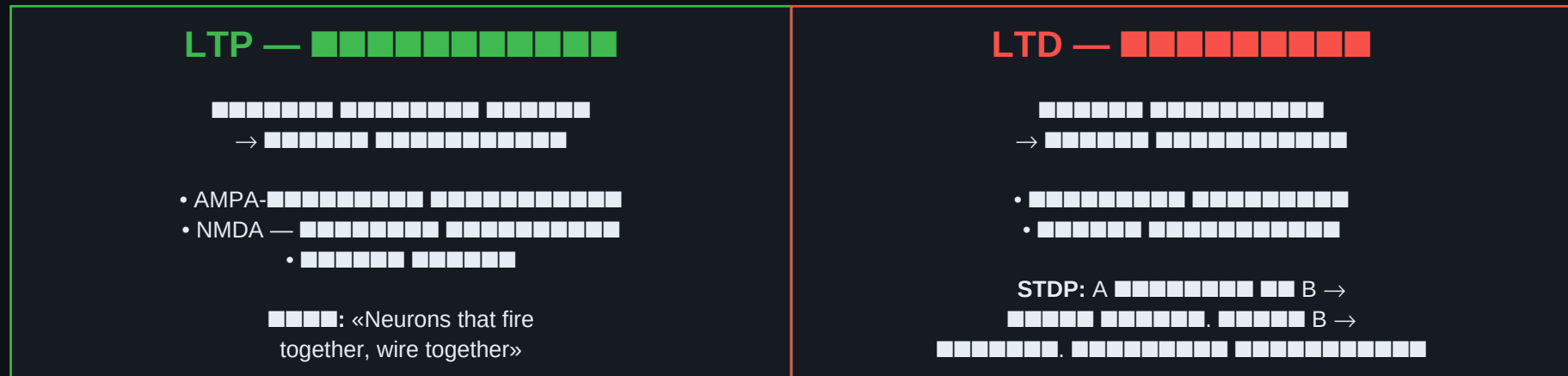
5. 5–9

1. 1990年12月，在北京市召开的“中国新闻改革国际研讨会”上，
 与会代表一致通过《中国新闻改革宣言》，指出：“中国新闻界正处
 于历史性的转折关头，中国新闻改革正在全面展开。”

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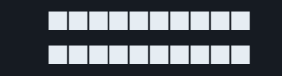
NREM — ■■■■■■ ■■■■
REM — ■■■■■■. ■■■ ■■■ —
■■■■■■■■■■ ■■■■■■■■ ■■■ 40%

LTP ■ LTD —



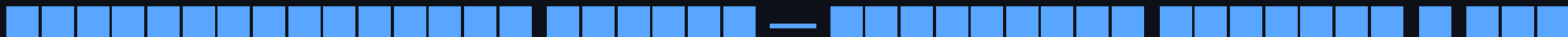
NMDA- — AND gate:

5.

1943	1958	1969	1986	2012	2017
					

XXXXXXXXXX — XXXXX XXXXXXX: XXXXXXXXXXXX XXXXXX XXXXXXX → XXXXXX. XXXXXXXXXXXX X, XXXX. XX XXXXXXX
XOR. XXXXXXXX — XXXXXXXXXXXXXXXX XXXXX + backpropagation (1986).

■■■■: ■■■■■■■■■■ (1958) → GPT-4 (2023) = 65 ■■■. ■■■■■ ■■■■■■■■■■ → ■■■■ = 8 ■■■. ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■.

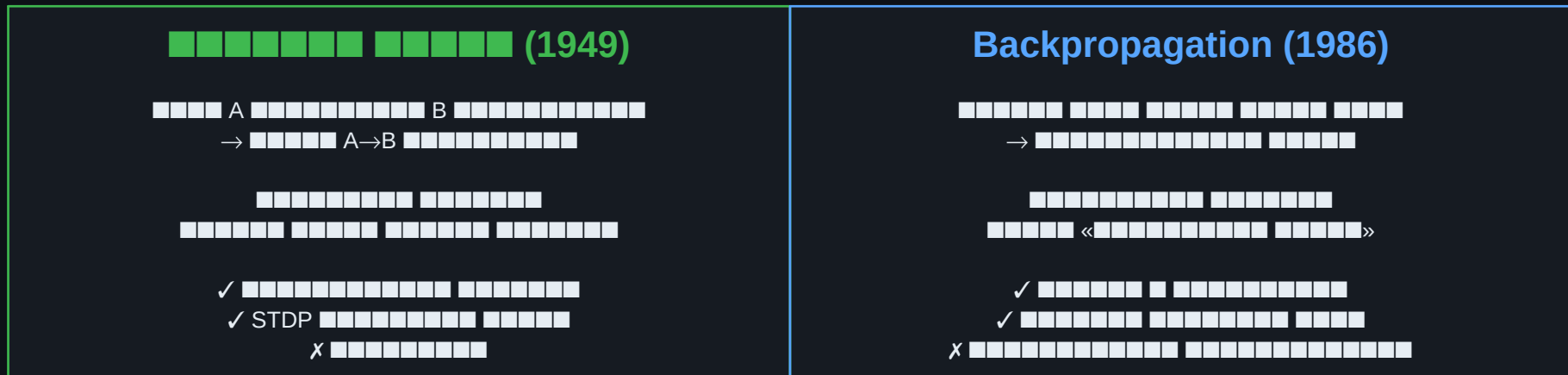


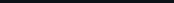
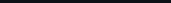
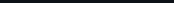
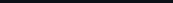
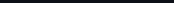
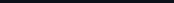
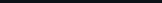
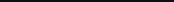
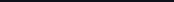
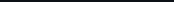
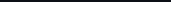
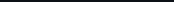
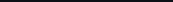
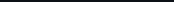
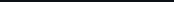
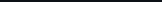
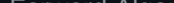
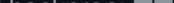
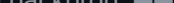
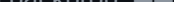
Year	System	Performance
1961	Stanford Cart	1000 ft/min. 1 min — 10–15 min. 32 min
1977	Tsukuba (1000 ft/min)	30 min/ft, 1000 ft/min
1986	Mercedes (1000 ft/min)	55–60 min/ft
1995	Navlab 5 (CMU)	4500 min, 98%
2005	Stanley (Stanford)	DARPA: 212 min 6 53 min
2014	Tesla Autopilot	
2017	Waymo	(1000 ft/min 4)

1000 ft/min — 1961 (Stanford Cart). 1000 ft/min — 2017 (Waymo). 56 min.

YOLO (1000 ft/min), U-Net (1000 ft/min), MonoDepth (1000 ft/min).

□□□□□□□□: □□□□ vs Backpropagation



Hinton  Forward-Forward Algorithm —                                                   

6. ■■■■ vs LLM — ■■■■

100 ■■■■
■■■■■■■■ (■■■■)

VS

1.8 ■■■■
■■■■■■■■■■ (GPT-4)

20 ■■
■■■

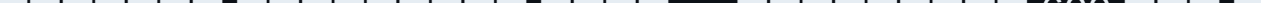
VS

500 000 ■■
■■■■■■■■ GPT-4

5

[illegible]




 600

$$\square\square\square\square\square\square\square\square\square\square\square\square\square\square - \square\square\square\square + \square\square\square\square + \square\square\square\square\square\square\square\square\square\square.$$

100 data scientists were asked to share the frameworks they used in their work. The results are shown below.

Framework	Percentage
TensorFlow	10%
Keras	20%
PyTorch	10%
Caffe	10%
MXNet	10%
Others	40%

PyTorch.

«If the brain were so simple we could understand it, we would be so simple we couldnt.» — Lyall Watson